Sections: B,C & D

Linear Algebra (MT-1004)

Homework#20

Instructor: Muhammad Ali

Question 1

Consider the following matrix A and the vector X_0:

$$A = \begin{pmatrix} -1 & 0 & 1 \\ 3 & 0 & -3 \\ 1 & 0 & -1 \end{pmatrix}, X_0 = \begin{pmatrix} 1 \\ 1 \\ 7 \end{pmatrix}.$$

Find

- $A^5 X_0$
- Find the general term of the discrete dynamical system $X_n = AX_{n-1}, n > 0$